

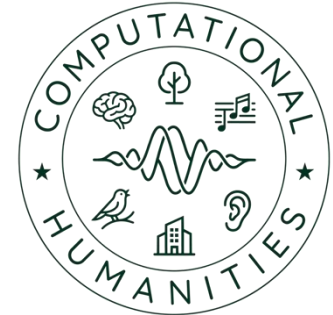
University of Bamberg



Master Project - Machine Listening (CH-Proj-M)

04 – Data in Machine Listening

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Lecture Structure



Week	Day	Project Phase	Topics
1	13.04.		Overview, master project organization, open project topics
2	20.04.	1. Literature research	Literature research strategies, reference management
3	27.04.		Basics: academic writing, required structure of the term paper
4	04.05. (P1)	2. Data acquisition (Dataset research, data recording)	Role of data and annotations in machine listening research, data protection, licences (CC etc.)
5	11.5.		Basics: audio editing and annotation (Audacity, Sonic Visualizer)

Data in Machine Listening

Challenges



- Paradigm shift
 - Earlier: knowledge-driven (features, classifier)
 - Today: data-driven (machine learning/deep learning)
- Same “problems” as for other domains (computer vision, natural language processing)
 - DL methods are “data-hungry”

Data in Machine Listening

Challenges



- Audio is “data-scarce” domain
 - Technical complexity
 - Audio is harder to scrape and extract than text or images
 - Slower and more expensive annotation
 - Labels require listening in real-time
 - Privacy risks
 - Strict regulations create legal hurdles for data sharing
 - Background noise requires costly filtering to ensure high-quality training signals

Data in Machine Listening

Challenges



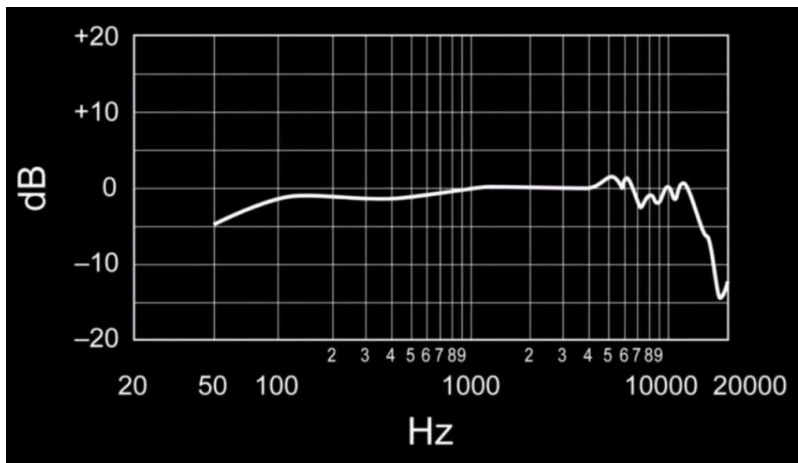
- Comparison to other domains

Domain	Common Large-Scale Dataset	Approximate Size
NLP	Common Crawl / C4	Trillions of tokens
Vision	ImageNet-21K / LAION	14 Million to 5 Billion+ images
Audio	LibriSpeech / AudioSet	1,000 hours of speech / 2 Million clips

Data in Machine Listening

Challenges

- Device-mismatch
 - High-quality devices



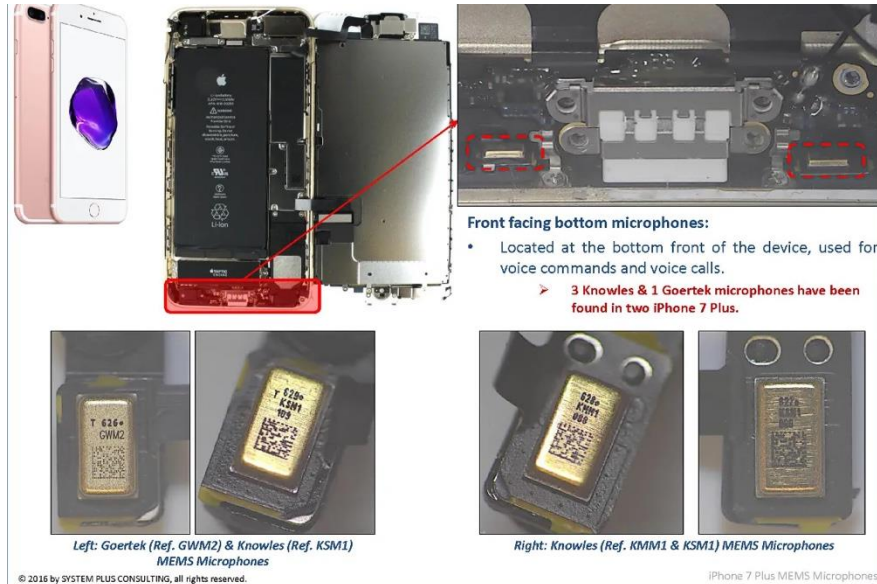
B1

B2

Data in Machine Listening

Challenges

- Low-quality devices (e.g. MEMS mike in cell phones)

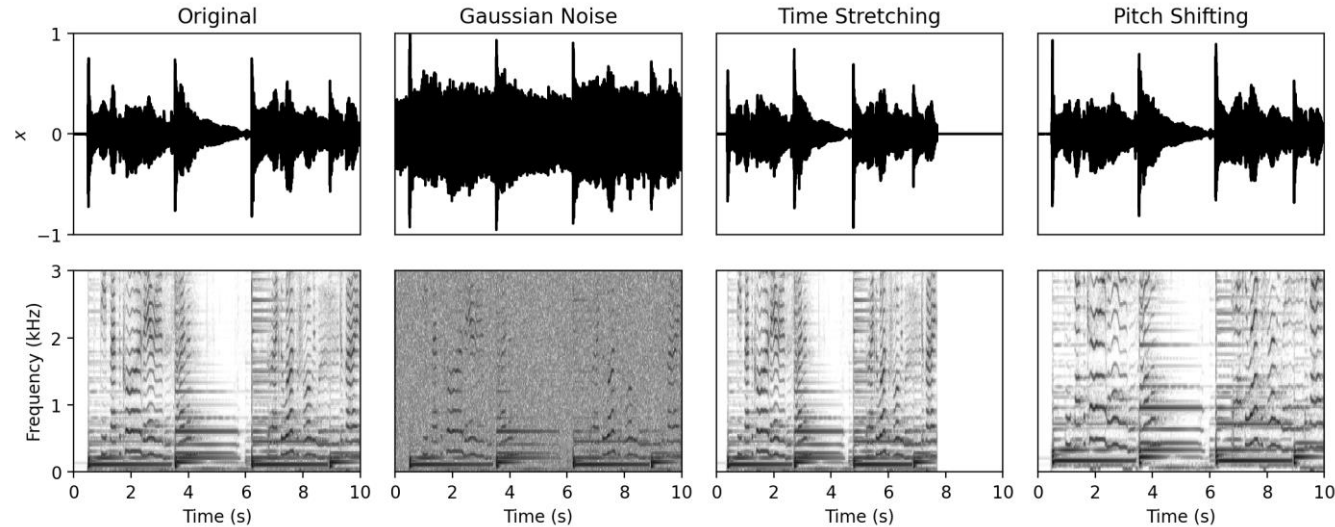


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Data in Machine Listening

Approaches against Data Scarcity

- Data augmentation



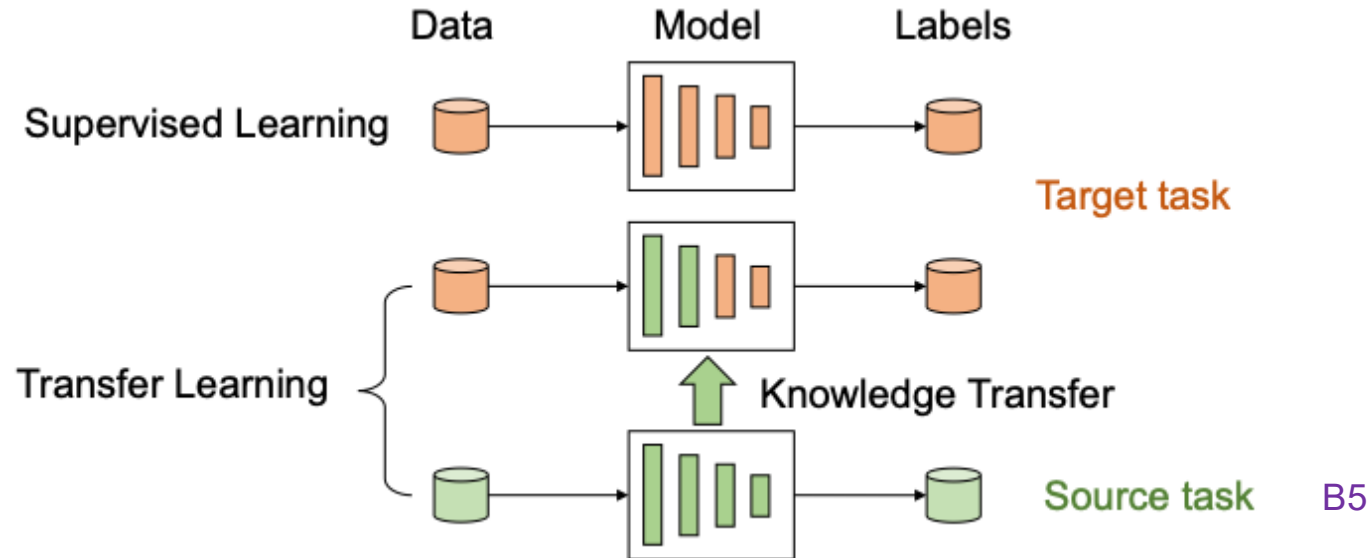
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Audiomentations (Python lib)

Data in Machine Listening

Approaches against Data Scarcity

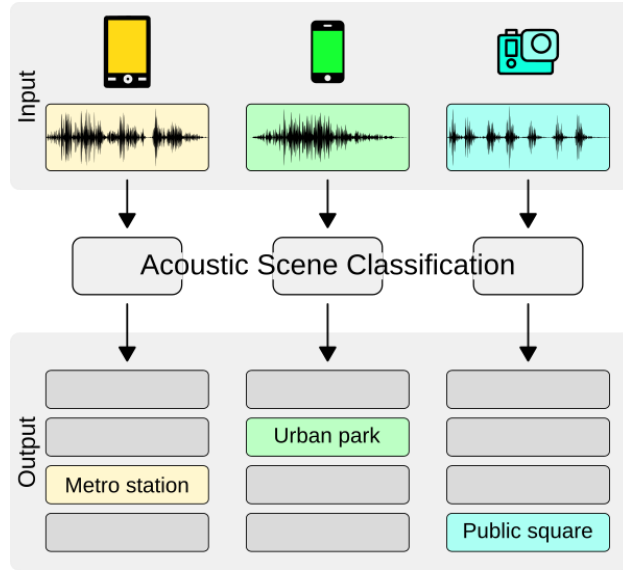
- Transfer learning



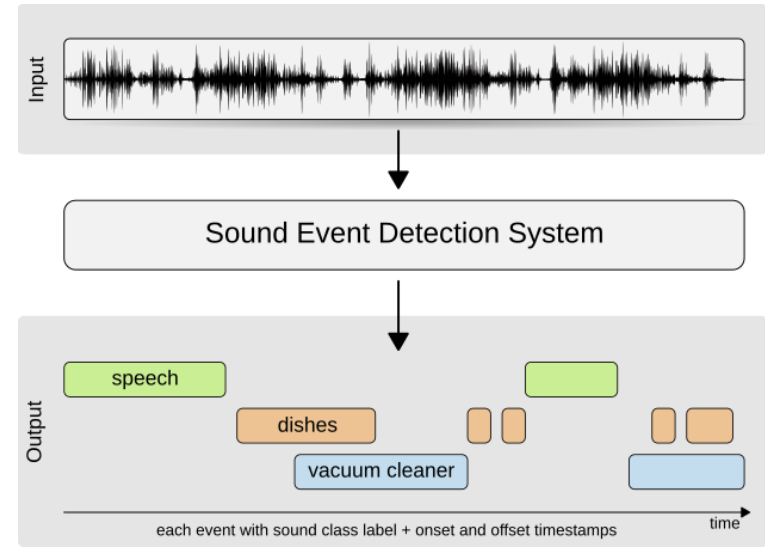
Data in Machine Listening

Types of Annotations

- Weak labels (clip-level, tagging) vs. strong labels (segment-label)



B6



B7

Data in Machine Listening

Open Data



- Reproducibility
 - Standardized pre-processing pipelines, model architectures, and training logs
- FAIR data principles
 - Optimizing audio datasets to be Findable, Accessible, Interoperable, and Reusable
- Good Scientific Practice (GSP)
 - Strict integrity in data handling—e.g. avoiding "data leakage" in train/test set

Data in Machine Listening

Data Protection



- Audio as Personal Data
 - Human speech contains biometric identifiers and PII (Personally Identifiable Information)
- Privacy-Enhancing Technologies (PETs)
 - Audio pseudonymization (pitch shifting/voice masking)
 - Differential privacy to prevent "speaker re-identification" from the latent space
- “Privacy-by-design” approaches

Data in Machine Listening

Licences



- Creative Commons (CC) Licence
 - CC0 (Public Domain Dedication) - "no rights reserved"
 - CC-BY (Attribution): Standard for open science, credit to the original creator is required
 - CC-BY-NC (Non-Commercial): Restricts usage to non-profit or academic purposes
 - CC-BY-SA (Share-Alike): Any derivative works (e.g., a processed version of your dataset) must be released under the same license

Data in Machine Listening

Licences



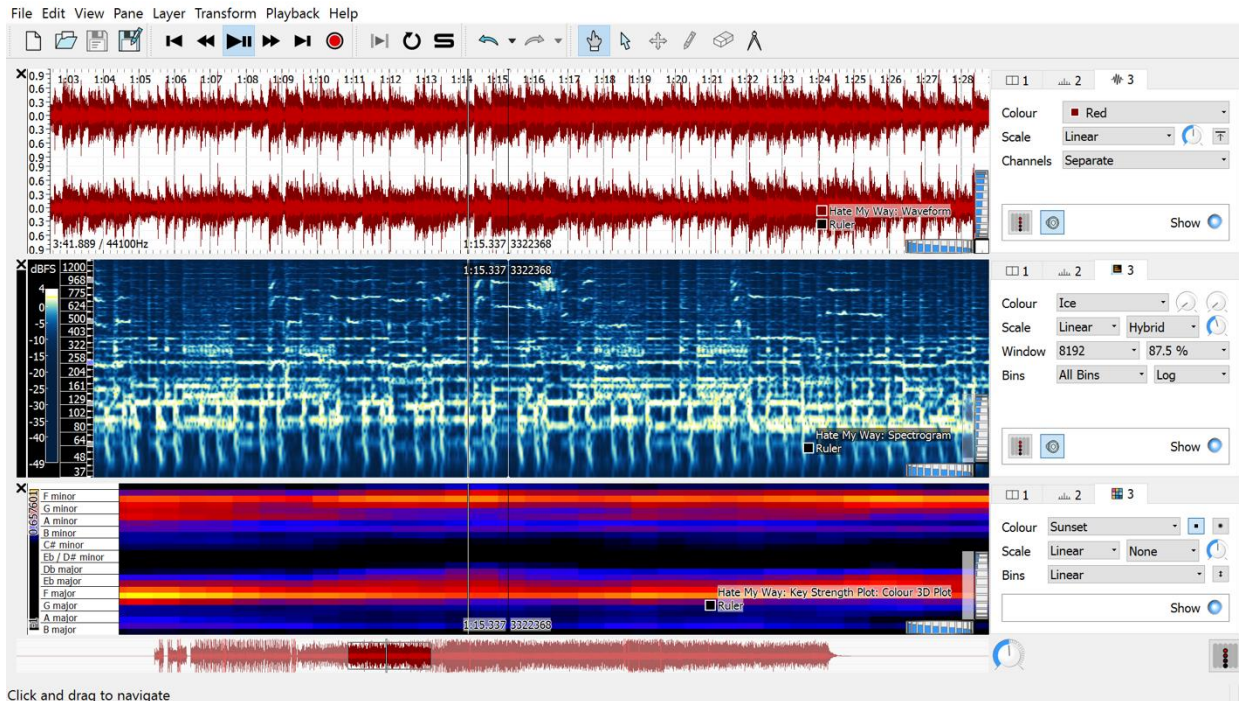
- MIT License
 - Permissive Software License
 - „Gold standard“ for scripts, scrapers, and processing
- "License Splitting"
 - Dual-license approach: CC-BY for the raw .wav files and MIT for the training/evaluation code

Data in Machine Listening

Sonic Visualiser



- [Download here](#)



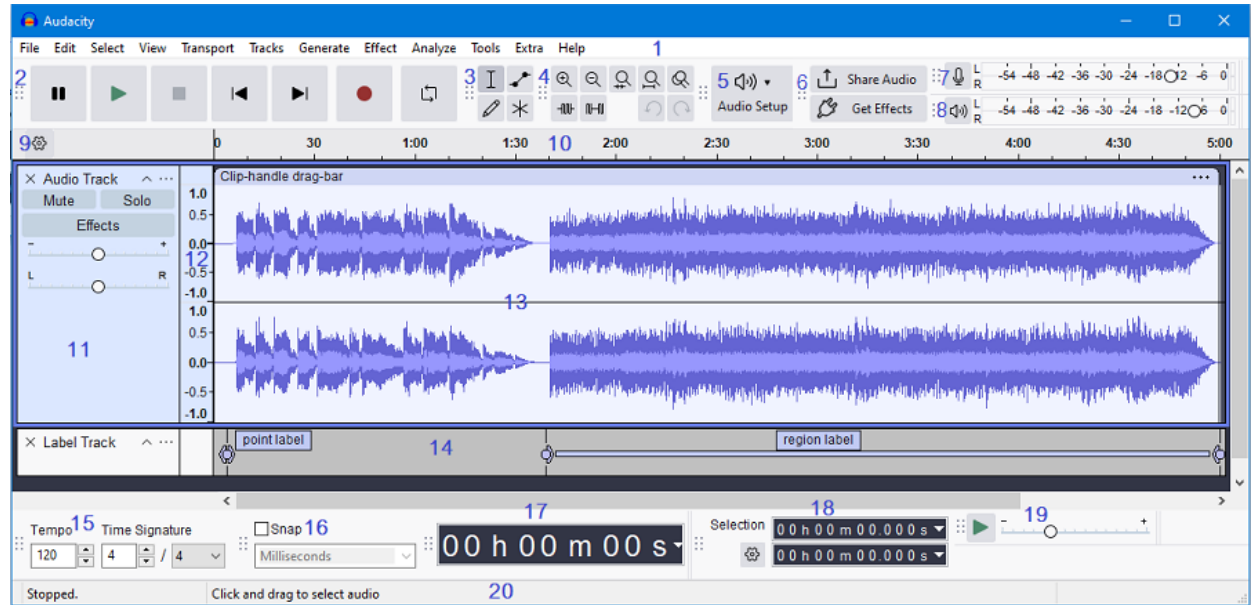
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Data in Machine Listening

Audacity



- [Download here](#)



B9

Presentation P2 (18.5.)

Required Structure

- 5 min presentation, 5 min Q/A
- Dataset(s) description
 - Metadata, data acquisition strategy (how recorded? which mics?)
 - Which classes?, #examples/class
 - Sample rate, audio format, file duration
 - Annotation (type of annotation, weak/strong)
 - Audio examples + Spectrogram examples
 - All team members shall present!

librosa.stft

Image Sources



B1: https://soundorp.com/wp-content/uploads/2025/01/InShot_20250120_113040009-1024x723.jpg

B2: <https://resources.rode.com/assets/ntsf1%20macro.PNG>

B3: https://www.eetasia.com/wp-content/uploads/sites/2/2020/04/iphone_7_mems_microphones_systemplusconsulting_cr.jpg

B4: (c) Jakob Abeßer

B5: (c) Jakob Abeßer

B6: https://dcase.community/images/tasks/challenge2020/task1_acoustic_scene_classification.png

B7: https://dcase.community/images/tasks/challenge2018/task4_sound_event_detection.png

B8: <https://www.sonicvisualiser.org/images/sv-3.0-win.png>

B9: https://manual.audacityteam.org/m/images/4/49/front_page_3_7_3_78_numbered.png

Thank you for your attention!

Any questions?



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